

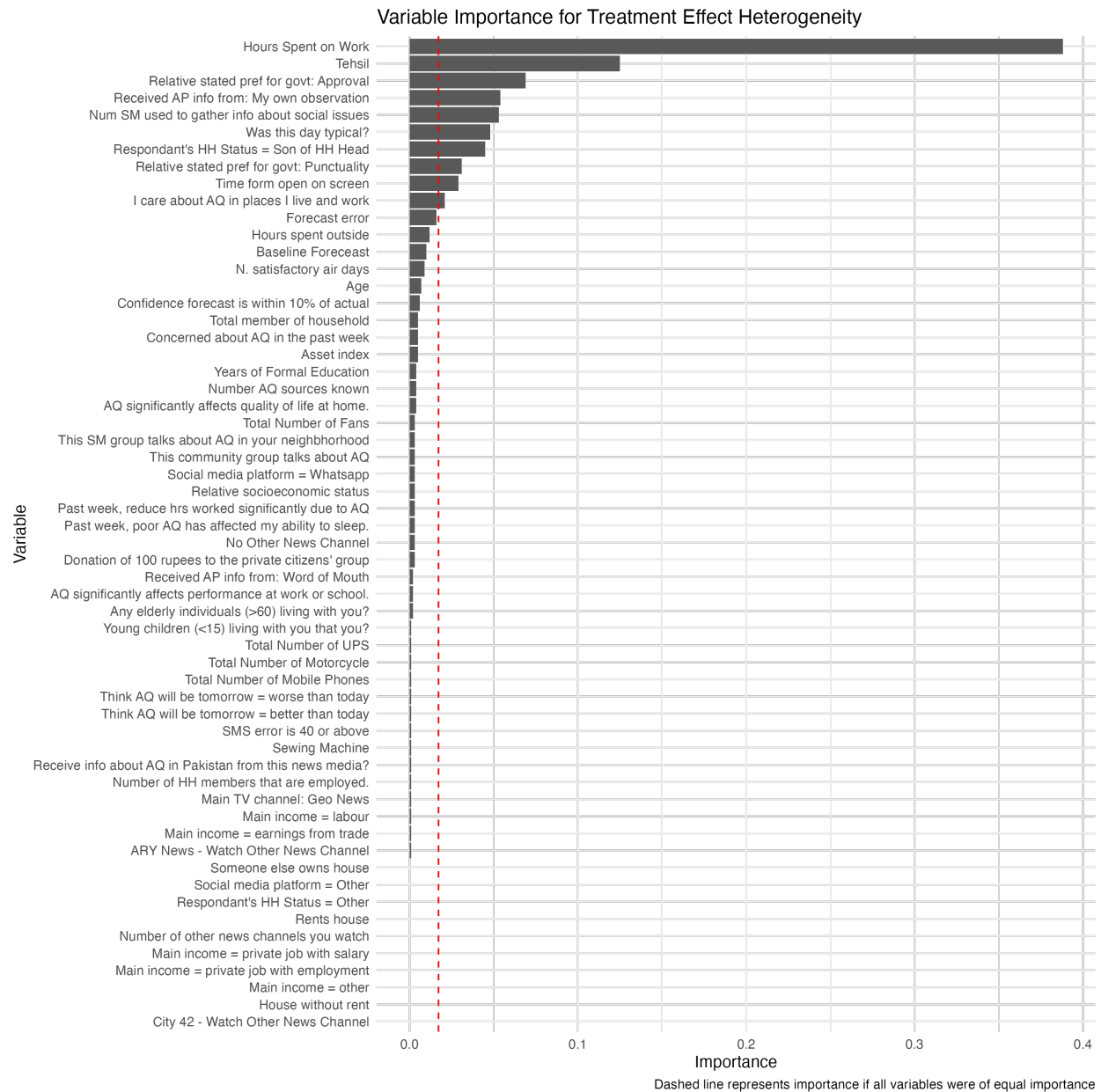
I analyzed heterogeneous treatment effects for air quality forecasting using causal forests.

Treatment = attribution to EPD (vs. PAQI).

Outcome = relative WTP, coded as  $WTP\_PAQI - WTP\_EPD$ .

All treatment effects are negative, meaning EPD attribution consistently reduces relative WTP (increases EPD preference). The analysis just tries to identify which household characteristics make this effect stronger.

## Variable Importance



Three variables account for the majority of treatment effect heterogeneity: hours spent on work (importance = 0.40), location/tehsil (importance = 0.10), and baseline relative government approval (importance = 0.05). Together, these explain 55% of variation in treatment effects.

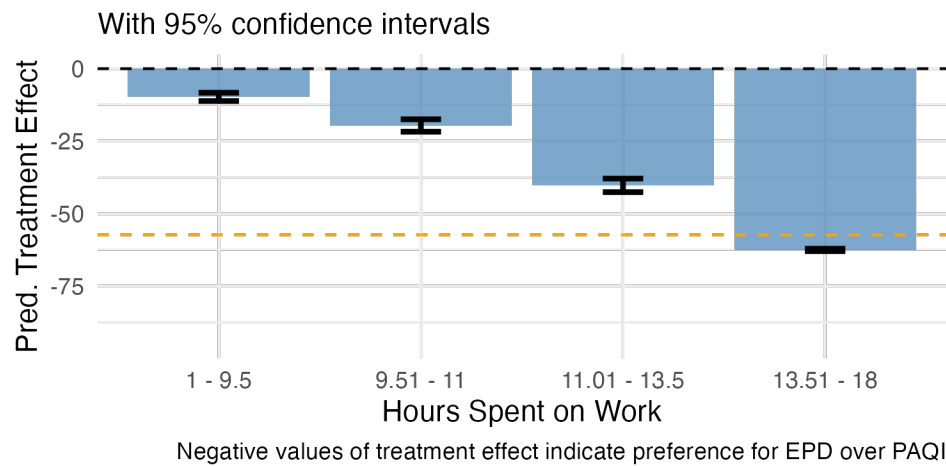
## Work Hours

Table 1: Best linear projection of work hours on treatment

| Term        | Estimate | Std..Error | t.value | P.Value   |
|-------------|----------|------------|---------|-----------|
| (Intercept) | 62.464   | 4.705      | 13.276  | < 2.2e-16 |
| Work Hours  | -8.260   | 0.398      | -20.751 | < 2.2e-16 |

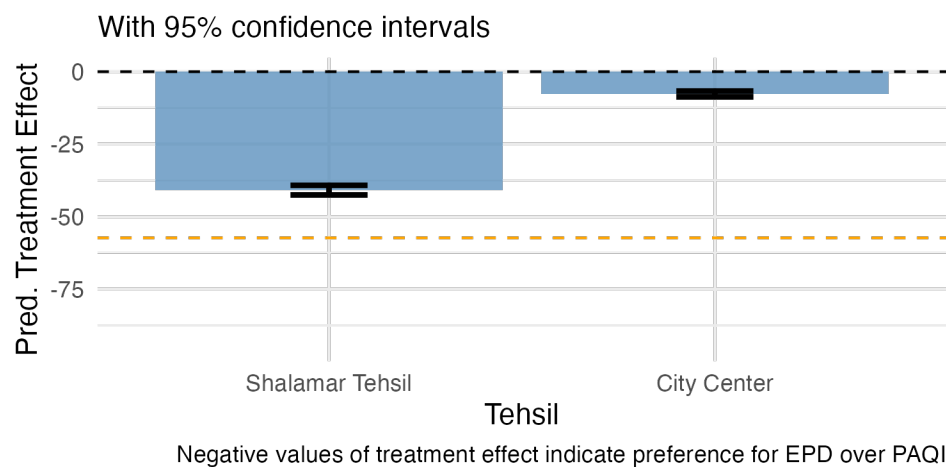
Each additional hour worked per week decreases the treatment effect by 8.3 PKR ( $p < 0.001$ ). This means households working more hours develop substantially stronger preferences for EPD attribution.

The dashed line is the (negative) average of willingness to pay over PAQI and EPD forecasting.



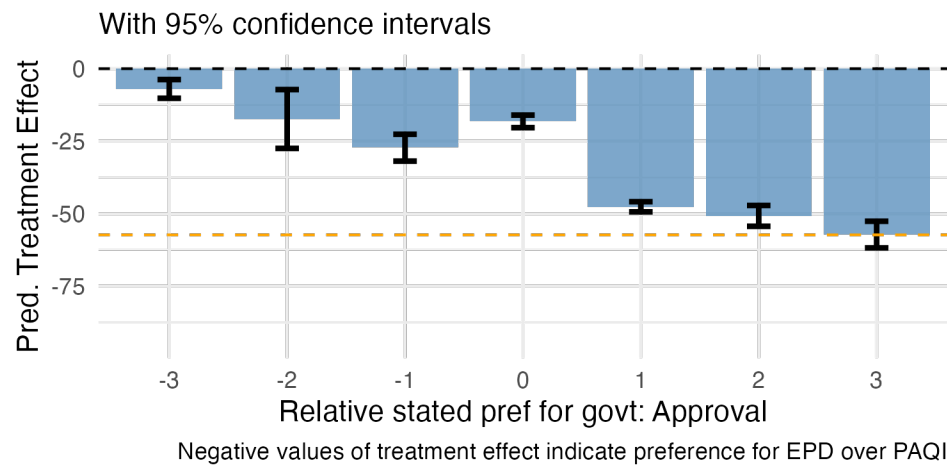
This plot also reflects that as work hours increase, treatment effect becomes more negative (more preference for EPD forecasting. )

## Location:



People living in Shalamar have a much more negative treatment effect.

### Approval index:



People with a positive approval for the government are more receptive to the EPD treatment.